Proving triangles congruent

ANSWER KEY

Core Concepts: Proving triangles congruent

1 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 104° ✓

b. 114°

c. 96°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 45^\circ) = 104^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 14

c. 12 ✓

d. 16

Substituting $x = 1$ into $y = 4x + 8$ gives $y = 4(1) + 8 = 12$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97° ✓

b. 107°

c. 89°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 12

b. 18

c. 16 ✓

d. 20

Substituting $x = 2$ into $y = 4x + 8$ gives $y = 4(2) + 8 = 16$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 90° ✓

b. 100°

c. 82°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 59^\circ) = 90^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 16

b. 22

c. 20 ✓

d. 24

Substituting $x = 3$ into $y = 4x + 8$ gives $y = 4(3) + 8 = 20$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 66^\circ) = 83^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24 ✓

d. 28

Substituting $x = 4$ into $y = 4x + 8$ gives $y = 4(4) + 8 = 24$.

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